

# HealthConnect Clinic, Week 7

## Project Summary

Data Analytics Track

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### 1. What I planned to test

The five KPIs (the four dashboard KPIs: prior no-show bracket, lead-time band, distance band, reminder sent, plus the cancellation-to-no-show ratio) against their source data, the two headline findings from Week 6 (lead-time risk and the distance-band downgrade) against new segments, the double-risk segment figures cited in the Week 6 report, the two questions Week 6 left open (whether distance overlaps with lead time, and how a double-risk pilot would run), and whether the DS track collaborator's model correctly encodes the double-risk finding it is documented as being built on.

### 2. What I actually tested

All five KPIs were reconciled or independently recomputed from the raw prepared dataset (four against the dashboard, KPI 5 against the Week 6 validation report). The lead-time finding was tested against two new segments (appointment type, age group). The distance-band downgrade was tested via a confidence-interval overlap check, and distance was tested for redundancy with lead time (correlation, in-band gaps, and a Logistic Regression with and without distance). The double-risk segment was reproduced from its stated definition and, as the mandatory HC-POD cross-track activity, the DS track collaborator's `double_risk_flag` feature was tested against the Data Analytics figures it claims to encode, under both a Random Forest and the exact Week 6 candidate model (a Logistic Regression).

### 3. Most important testing results

All five KPIs reproduced exactly. The lead-time finding held in every subgroup of both new segments. The distance-band downgrade was reconfirmed. Distance is not redundant with lead time (correlation  $-0.008$ ); the 15+ km gap appears inside every lead-time band at about +7 points, but it is modest. The double-risk reproduction failed on the first attempt and was root-caused: the correct definition uses a lead-time threshold of 31+ days, not 46+ as initially assumed. The DS track collaborator's coded feature uses the narrower 46+ threshold.

### 4. Issues or weaknesses identified

Three real issues: (1) an internal inconsistency in how the double-risk segment was first reproduced versus how it was originally defined; (2) a documentation-to-code mismatch in the DS track collaborator's code, where the `double_risk_flag` threshold ( $\geq 46$  days) does not match the Data Analytics finding it is documented as representing ( $\geq 31$  days), silently excluding 110 at-risk patients from the flag; (3) the Week 6 Decision Support text also states the flag as 46+ days, which is superseded by the validated  $\geq 31$  definition.

### 5. Improvements/refinements made

The double-risk segment definition was corrected and reconfirmed against the raw data. A visual refinement for the distance-band chart was designed (grouping the three overlapping bands, isolating the 15+ km band, adding a CI-overlap annotation), with a mockup, to bring the chart in line with the validated finding. Implementation in the live dashboard is still pending. KPI 5 coverage and the distance redundancy check were added to the test suite.

### 6. Retesting results

The corrected double-risk segment was retested via bootstrap resampling and found stable (95% CI [67.4%, 79.0%]). The corrected threshold was retested on the Random Forest (0.6661 vs 0.6662 ROC-AUC) and on the exact Week 6 candidate Logistic Regression (0.6819 for  $\geq 46$  vs 0.6821 for  $\geq 31$  with the Week 6 method; paired difference  $+0.0001$  over  $5 \times 5$  repeated CV), confirming the fix is a documentation/consistency correction, not a performance-affecting one.

### 7. Track(s) collaborated with

The DS track collaborator.

## 8. What was tested collaboratively

Whether the DS track collaborator's `double_risk_flag` feature engineering correctly reproduces the Data Analytics-validated double-risk finding it cites as its rationale.

## 9. What changed as a result

A documented, evidence-based correction to the `double_risk_flag` threshold was produced, along with confirmation, under two model types, that the original threshold was not masking a model performance problem. The  $\geq 31$  definition was kept, and the matching 46+ wording in the Week 6 text was flagged as superseded.

## 10. Key findings or validation outcomes

Booking lead time remains the strongest, most validated predictor of no-show risk. The double-risk segment (long lead time + repeated prior no-shows) is a small, stable, high-risk group worth targeting directly. Distance to clinic should be treated as a threshold effect (15+ km), which is real inside every lead-time band but modest. Long lead-time bookings fail mostly through silent no-shows, not cancellations (ratio falls from about 0.20 to 0.08).

## 11. Major challenges encountered

The double-risk segment's exact definition was ambiguous enough on first pass to produce an incorrect reproduction (2.46% of bookings / 82.1% risk instead of the correct 4.66% / 73.0%), which required tracing back through the lead-time band boundaries to resolve. The distance checks were limited by small 15+ km groups (118 to 259 patients per lead-time band).

## 12. Important decisions made

Corrected the `double_risk_flag` threshold to  $\geq 31$  days rather than leave the discrepancy undocumented, and kept it at  $\geq 31$  after the exact-candidate Logistic Regression retest, because it covers 233 high-risk patients at 73.0% against 123 at 82.1% for  $\geq 46$  and the model results are identical. Treated the call pilot as a design only, since it cannot be run on this dataset.

## 13. Remaining limitations

The distance-band chart refinement has been designed but not yet applied to the live Power BI dashboard. The `reminder_sent` and `reminder_channel` findings were not subjected to the same depth of testing this week and remain lower-confidence. The in-band distance test uses simple intervals and small subgroups. The flag test is cross-validated on one dataset with no held-out check, and model counts (4,737 rows) differ from DA counts (5,000 rows) because cancelled bookings are excluded from modelling. Dashboard slicers did not exist and their testing is pending.

## 14. Remaining issues or dependencies

The corrected `double_risk_flag` threshold has been validated in a standalone retest but has not yet been merged into the DS track collaborator's `train_improved.py` pipeline. The call pilot depends on clinic agreement and enough bookings (about 14,600 for a 10-point effect).

## 15. Contribution to the overall HealthConnect project

This week's testing confirmed the reliability of the Data Analytics track's dashboard and core findings, closed two open Week 6 questions, and caught a real cross-track inconsistency before it could carry an incorrect assumption into Week 8 final integration, directly supporting the project's end-to-end validation goal.

## 16. What must be completed before Week 8

Apply the distance-band chart refinement to the live dashboard and re-export it; the DS track collaborator to merge the corrected `double_risk_flag` threshold into their codebase; agree the pilot design with clinic staff; add a held-out or new-data check of the flag threshold; complete final packaging (Drive folder, GitHub README update) and submit.